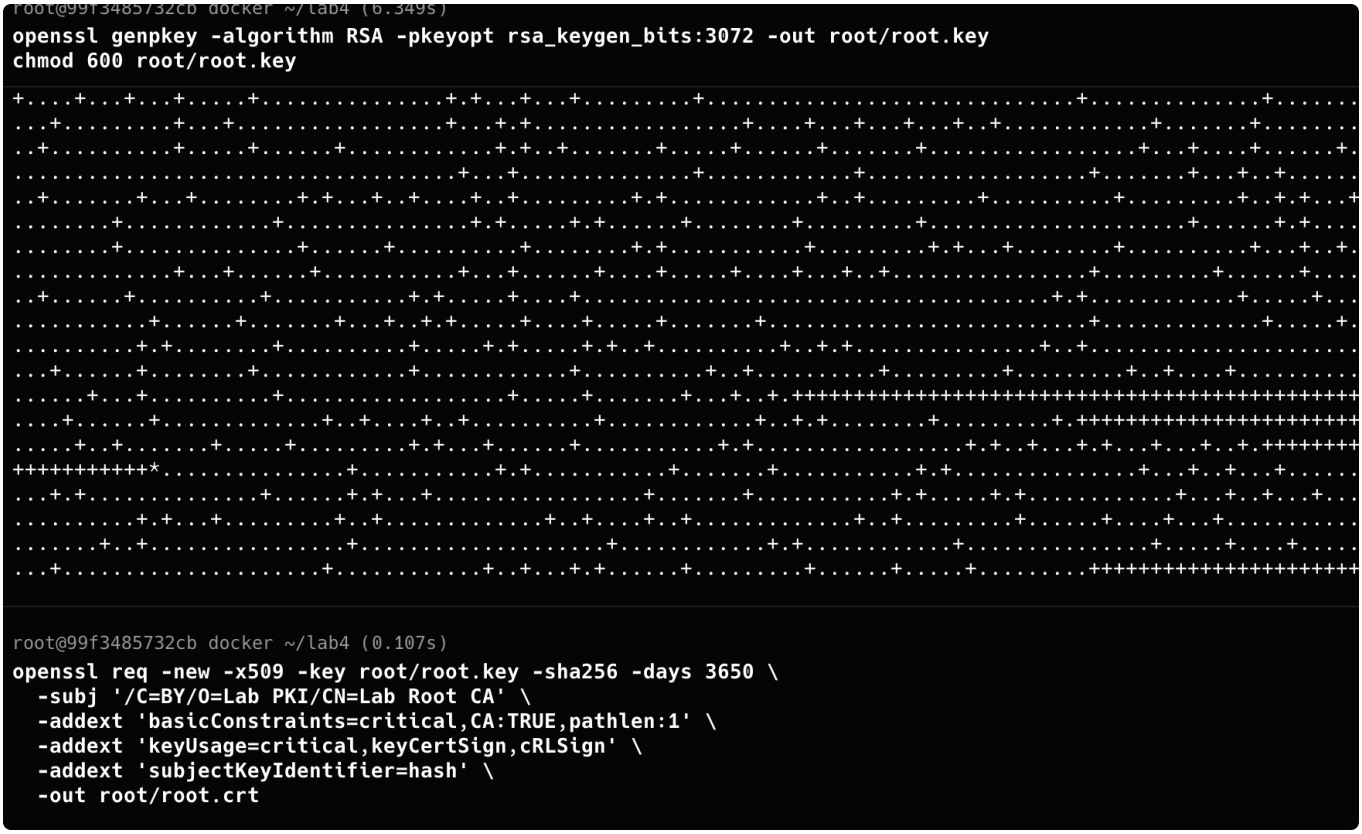
1. Корневой CA

```
openssl genpkey -algorithm RSA -pkeyopt rsa_keygen_bits:3072 -out
root/root.key
chmod 600 root/root.key

openssl req -new -x509 -key root/root.key -sha256 -days 3650 \
  -subj '/C=BY/O=Lab PKI/CN=Lab Root CA' \
  -addext 'basicConstraints=critical,CA:TRUE,pathlen:1' \
  -addext 'keyUsage=critical,keyCertSign,cRLSign' \
  -addext 'subjectKeyIdentifier=hash' \
  -out root/root.crt
```



проверка:

```
openssl x509 -in root/root.crt -noout -subject -issuer -ext basicConstraints
```

```
root@99f3485732cb docker ~/lab4 (0.09s)
openssl x509 -in root/root.crt -noout -subject -issuer -ext basicConstraints
subject=C = BY, O = Lab PKI, CN = Lab Root CA
issuer=C = BY, O = Lab PKI, CN = Lab Root CA
X509v3 Basic Constraints: critical
    CA:TRUE, pathlen:1
```

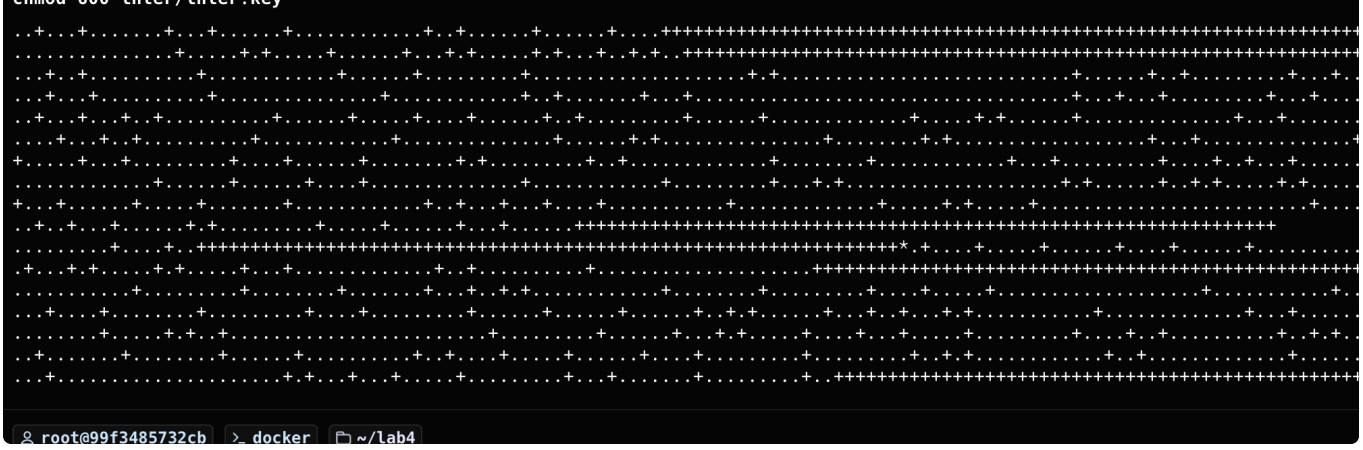
2. Промежуточный CA

```
cat > inter/inter.ext <<'EOF'
basicConstraints=critical,CA:TRUE,pathlen:0
keyUsage=critical,keyCertSign,cRLSign
subjectKeyIdentifier=hash
authorityKeyIdentifier=keyid:always
EOF

openssl genpkey -algorithm RSA -pkeyopt rsa_keygen_bits:3072 -out
inter/inter.key
chmod 600 inter/inter.key
```

```
root@99f3485732cb docker ~/lab4 (0.056s)
cat > inter/inter.ext <<'EOF'
basicConstraints=critical,CA:TRUE,pathlen:0
keyUsage=critical,keyCertSign,cRLSign
subjectKeyIdentifier=hash
authorityKeyIdentifier=keyid:always
EOF

root@99f3485732cb docker ~/lab4 (2.462s)
openssl genpkey -algorithm RSA -pkeyopt rsa_keygen_bits:3072 -out inter/inter.key
chmod 600 inter/inter.key
```



```
root@99f3485732cb  docker  ~/lab4
```

```
openssl req -new -key inter/inter.key \
  -subj '/C=BY/O=Lab PKI/CN=Lab Intermediate CA' \
  -out inter/inter.csr

openssl x509 -req -in inter/inter.csr \
  -CA root/root.crt -CAkey root/root.key -CAcreateserial \
```

```
  -sha256 -days 1825 -extfile inter/inter.ext \
  -out inter/inter.crt
```

```
root@99f3485732cb docker ~/lab4 (0.053s)
openssl req -new -key inter/inter.key \
  -subj '/C=BY/O=Lab PKI/CN=Lab Intermediate CA' \
  -out inter/inter.csr
```

```
root@99f3485732cb docker ~/lab4 (0.156s)
openssl x509 -req -in inter/inter.csr \
  -CA root/root.crt -CAkey root/root.key -CAcreateserial \
  -sha256 -days 1825 -extfile inter/inter.ext \
  -out inter/inter.crt
Certificate request self-signature ok
subject=C = BY, O = Lab PKI, CN = Lab Intermediate CA
```

проверка:

```
openssl x509 -in inter/inter.crt -noout -subject -issuer -ext
basicConstraints,keyUsage
```

```
root@99f3485732cb docker ~/lab4 (0.075s)
openssl x509 -in inter/inter.crt -noout -subject -issuer -ext basicConstraints,keyUsage
subject=C = BY, O = Lab PKI, CN = Lab Intermediate CA
issuer=C = BY, O = Lab PKI, CN = Lab Root CA
X509v3 Basic Constraints: critical
    CA:TRUE, pathlen:0
X509v3 Key Usage: critical
    Certificate Sign, CRL Sign
```

3. База промежуточного CA

```
cd ~/lab4/inter
touch index.txt
echo 01 > crlnumber
echo 1000 > serial
cd ~/lab4

cat > inter/ca.cnf <<EOF
[ ca ]
default_ca = CA_default

[ CA_default ]
```

```
dir              = $HOME/lab4/inter
database         = \$dir/index.txt
new_certs_dir    = \$dir
serial           = \$dir/serial
crlnumber        = \$dir/crlnumber
certificate      = \$dir/inter.crt
private_key      = \$dir/inter.key
default_md       = sha256
default_days     = 365
default_crl_days = 30
unique_subject   = no
copy_extensions  = none
policy           = policy_any

[ policy_any ]
commonName       = supplied
EOF
```

```
root@99f3485732cb docker ~/lab4 (0.038s)
cd ~/lab4/inter
touch index.txt
echo 01 > crlnumber
echo 1000 > serial
cd ~/lab4
```

```
root@99f3485732cb docker ~/lab4 (0.037s)
cat > inter/ca.cnf <<EOF
[ ca ]
default_ca = CA_default

[ CA_default ]
dir               = $HOME/lab4/inter
database          = \$dir/index.txt
new_certs_dir     = \$dir
serial            = \$dir/serial
crlnumber         = \$dir/crlnumber
certificate       = \$dir/inter.crt
private_key       = \$dir/inter.key
default_md        = sha256
default_days      = 365
default_crl_days  = 30
unique_subject    = no
copy_extensions   = none
policy            = policy_any

[ policy_any ]
commonName        = supplied
EOF
```

проверка:

```
grep -E '^(dir|database|certificate|private_key)' inter/ca.cnf
```

```
root@99f3485732cb docker ~/lab4 (0.035s)
grep -E '^(dir|database|certificate|private_key)' inter/ca.cnf
dir               = /root/lab4/inter
database          = $dir/index.txt
certificate       = $dir/inter.crt
private_key       = $dir/inter.key
```

4. Серверный сертификат: RSA-2048, SAN localhost

```
cat > server/server.ext <<'EOF'
basicConstraints=critical,CA:FALSE
keyUsage=critical,digitalSignature
extendedKeyUsage=serverAuth
subjectKeyIdentifier=hash
authorityKeyIdentifier=keyid:always
subjectAltName=DNS:localhost,IP:127.0.0.1
EOF

openssl genpkey -algorithm RSA -pkeyopt rsa_keygen_bits:2048 -out
server/server.key
chmod 600 server/server.key
```

```
root@99f3485732cb docker ~/lab4 (0.413s)
cat > server/server.ext <<'EOF'
basicConstraints=critical,CA:FALSE
keyUsage=critical,digitalSignature
extendedKeyUsage=serverAuth
subjectKeyIdentifier=hash
authorityKeyIdentifier=keyid:always
subjectAltName=DNS:localhost,IP:127.0.0.1
EOF

openssl genpkey -algorithm RSA -pkeyopt rsa_keygen_bits:2048 -out server/server.key
chmod 600 server/server.key
..+.........+..+...................+.....+...+.+.....+.......+...+..........+.+.....+...+.+...+++++
+++++*..++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++*..+.....+.........+.....
++++++++++++++++++++++++
.........+..+.+.....+.+.....+.+...+..+.......+..+.+.........+.........+..+.......+..+......+......
++++++++++++++++++++++++++++++++++++++++++++++++++*.+........+......++++++++++++++++++++++++++++++++
.+...........+.......+........+......+....+..........+.+..........+.......+...+.....+.........+.
++++++++++++++++++++++++++++++++++++++++++
```

```
openssl req -new -key server/server.key -out server/server.csr -subj
"/C=BY/O=Lab PKI/CN=localhost"

openssl ca -config inter/ca.cnf -batch \
  -in server/server.csr -extfile server/server.ext \
  -notext -out server/server.crt
```

```
root@99f3485732cb docker ~/lab4 (0.042s)
openssl req -new -key server/server.key -out server/server.csr -subj "/C=BY/O=Lab PKI/CN=localhos
```

```
root@99f3485732cb docker ~/lab4 (0.112s)
openssl ca -config inter/ca.cnf -batch \
  -in server/server.csr -extfile server/server.ext \
  -notext -out server/server.crt
Using configuration from inter/ca.cnf
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
countryName           :PRINTABLE:'BY'
organizationName      :ASN.1 12:'Lab PKI'
commonName            :ASN.1 12:'localhost'
Certificate is to be certified until Nov 12 08:57:28 2027 GMT (365 days)

Write out database with 1 new entries
Database updated
```

5. Проверка цепочки

```
openssl verify –CAfile root/root.crt –untrusted inter/inter.crt
server/server.crt

openssl verify –CAfile root/root.crt server/server.crt
```

```
root@99f3485732cb docker ~/lab4 (0.103s)
openssl verify -CAfile root/root.crt -untrusted inter/inter.crt server/server.crt
server/server.crt: OK
```

```
root@99f3485732cb docker ~/lab4 (0.035s)
openssl verify -CAfile root/root.crt server/server.crt
CN = localhost
error 20 at 0 depth lookup: unable to get local issuer certificate
error server/server.crt: verification failed
```

6. nginx

```
cat server/server.crt inter/inter.crt > server/fullchain.pem

cat > /etc/nginx/conf.d/lab4.conf <<'EOF'
server {
    listen 8443 ssl;
    server_name localhost;

    ssl_certificate     /root/lab4/server/fullchain.pem;
    ssl_certificate_key /root/lab4/server/server.key;

    ssl_protocols TLSv1.3;
```

```
    ssl_conf_command Ciphersuites TLS_AES_256_GCM_SHA384;
    ssl_ecdh_curve P-256;
    ssl_session_tickets on;

    location / { return 200 'tls13 ok'; }
}
EOF

nginx -t

nginx -s reload 2>/dev/null || nginx
```

```
root@99f3485732cb docker ~/lab4 (0.046s)
cat server/server.crt inter/inter.crt > server/fullchain.pem
```

```
root@99f3485732cb docker ~/lab4 (0.031s)
cat > /etc/nginx/conf.d/lab4.conf <<'EOF'
server {
    listen 8443 ssl;
    server_name localhost;

    ssl_certificate     /root/lab4/server/fullchain.pem;
    ssl_certificate_key /root/lab4/server/server.key;

    ssl_protocols TLSv1.3;
    ssl_conf_command Ciphersuites TLS_AES_256_GCM_SHA384;
    ssl_ecdh_curve P-256;
    ssl_session_tickets on;

    location / { return 200 'tls13 ok'; }
}
EOF
```

```
root@99f3485732cb docker ~/lab4 (0.046s)
nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
```

```
root@99f3485732cb docker ~/lab4 (0.053s)
nginx -s reload 2>/dev/null || nginx
```

7. Провека curl

```
curl -v --cacert root/root.crt https://localhost:8443/
```

```
root@99f3485732cb docker ~/lab4 (0.153s)
curl -v --cacert root/root.crt https://localhost:8443/
* Host localhost:8443 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
*   Trying [::1]:8443...
* connect to ::1 port 8443 from ::1 port 33032 failed: Connection refused
*   Trying 127.0.0.1:8443...
* Connected to localhost (127.0.0.1) port 8443
* ALPN: curl offers h2,http/1.1
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
*  CAfile: root/root.crt
*  CApath: /etc/ssl/certs
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.3 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.3 (IN), TLS handshake, Encrypted Extensions (8):
* TLSv1.3 (IN), TLS handshake, Certificate (11):
* TLSv1.3 (IN), TLS handshake, CERT verify (15):
* TLSv1.3 (IN), TLS handshake, Finished (20):
* TLSv1.3 (OUT), TLS handshake, Finished (20):
* SSL connection using TLSv1.3 / TLS_AES_256_GCM_SHA384 / prime256v1 / RSASSA-PSS
* ALPN: server accepted http/1.1
* Server certificate:
*  subject: CN=localhost
*  start date: Nov 12 08:57:28 2026 GMT
*  expire date: Nov 12 08:57:28 2027 GMT
*  subjectAltName: host "localhost" matched cert's "localhost"
*  issuer: C=BY; O=Lab PKI; CN=Lab Intermediate CA
*  SSL certificate verify ok.
*   Certificate level 0: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
*   Certificate level 1: Public key type RSA (3072/128 Bits/secBits), signed using sha256WithRSAEncryption
*   Certificate level 2: Public key type RSA (3072/128 Bits/secBits), signed using sha256WithRSAEncryption
* using HTTP/1.x
> GET / HTTP/1.1
```

проверка:

```
curl https://localhost:8443/;
```

```
root@99f3485732cb docker ~/lab4 (0.284s)
curl https://localhost:8443/
curl: (60) SSL certificate problem: unable to get local issuer certificate
More details here: https://curl.se/docs/sslcerts.html

curl failed to verify the legitimacy of the server and therefore could not
establish a secure connection to it. To learn more about this situation and
how to fix it, please visit the web page mentioned above.
```

8. Захват рукопожатия и журнал ключей

1 терминал:

```
docker exec -it lab4 tcpdump -i lo -w /root/lab4/client/handshake.pcap 'tcp
port 8443'
```

```
base ~ (46s)
docker exec -it lab4 tcpdump -i lo -w /root/lab4/client/handshake.pcap 'tcp port 8443'
tcpdump: listening on lo, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

2 терминал:

```
docker exec -it lab4 bash -lc '
  cd /root/lab4/client &&
  SSLKEYLOGFILE=$HOME/lab4/client/keys.log \
    curl -sS --cacert ../root/root.crt https://localhost:8443/'
```

```
base ~ (0.202s)
docker exec -it lab4 bash -lc '
  cd /root/lab4/client &&
  SSLKEYLOGFILE=$HOME/lab4/client/keys.log \
    curl -sS --cacert ../root/root.crt https://localhost:8443/'
tls13 ok%
```

сервер(docker):

```
cd ~/lab4/client
wc -l keys.log
cut -d' ' -f1 keys.log | sort -u
```

```
root@99f3485732cb docker ~/lab4/client (0.037s)
cd ~/lab4/client
wc -l keys.log
cut -d' ' -f1 keys.log | sort -u

10 keys.log
CLIENT_HANDSHAKE_TRAFFIC_SECRET
CLIENT_TRAFFIC_SECRET_0
EXPORTER_SECRET
SERVER_HANDSHAKE_TRAFFIC_SECRET
SERVER_TRAFFIC_SECRET_0
```

9. Анализ в Wireshark

```
cd ~/lab4
tshark -r client/handshake.pcap -Y tls \
  -T fields -e frame.number -e tls.record.content_type -e tls.handshake.type

tshark -r client/handshake.pcap -o tls.keylog_file:client/keys.log \
  -Y tls.handshake -T fields -e frame.number -e tls.handshake.type
```

```
root@99f3485732cb docker ~/lab4 (0.245s)
cd ~/lab4
tshark -r client/handshake.pcap -Y tls \
  -T fields -e frame.number -e tls.record.content_type -e tls.handshake.type

Running as user "root" and group "root". This could be dangerous.
6       22      1
8       22,20   2
10      20
11
12
13
14
15
```

```
root@99f3485732cb docker ~/lab4 (0.305s)
tshark -r client/handshake.pcap -o tls.keylog_file:client/keys.log \
  -Y tls.handshake -T fields -e frame.number -e tls.handshake.type

Running as user "root" and group "root". This could be dangerous.
6       1
8       2,8,11,15,20
10      20
11      4
12      4
```

проверка:

```
tshark -r handshake.pcap -o tls.keylog_file:keys.log \
  -Y 'tls.handshake.type == 2' \
  -T fields -e tls.handshake.ciphersuite -e
tls.handshake.extensions_key_share_group

tshark -r handshake.pcap -o tls.keylog_file:keys.log \
  -Y 'tls.handshake.type == 1' \
  -T fields -e tls.handshake.extensions_server_name

tshark -r handshake.pcap -o tls.keylog_file:keys.log \
  -Y 'tls.handshake.type == 11' -V | grep -E 'dNSName|iPAddress'

tshark -r handshake.pcap -o tls.keylog_file:keys.log \
  -Y 'tls.handshake.type == 15' \
  -T fields -e tls.handshake.sig_hash_alg
```

```
root@99f3485732cb docker ~/lab4/client (0.692s)
tshark -r handshake.pcap -o tls.keylog_file:keys.log \
  -Y 'tls.handshake.type == 2' \
  -T fields -e tls.handshake.ciphersuite -e tls.handshake.extensions_key_share_group

tshark -r handshake.pcap -o tls.keylog_file:keys.log \
  -Y 'tls.handshake.type == 1' \
  -T fields -e tls.handshake.extensions_server_name

tshark -r handshake.pcap -o tls.keylog_file:keys.log \
  -Y 'tls.handshake.type == 11' -V | grep -E 'dNSName|iPAddress'

tshark -r handshake.pcap -o tls.keylog_file:keys.log \
  -Y 'tls.handshake.type == 15' \
  -T fields -e tls.handshake.sig_hash_alg
Running as user "root" and group "root". This could be dangerous.
0x1302  23
Running as user "root" and group "root". This could be dangerous.
localhost
Running as user "root" and group "root". This could be dangerous.
                                    GeneralName: dNSName (2)
                                        dNSName: localhost
                                    GeneralName: iPAddress (7)
                                        iPAddress: 127.0.0.1
Running as user "root" and group "root". This could be dangerous.
0x0804
```

10. Индивидуальное задание

```
mkdir -p ~/lab4/pathlen/root ~/lab4/pathlen/inter
cd ~/lab4/pathlen

openssl genpkey -algorithm RSA -pkeyopt rsa_keygen_bits:3072 -out
root/root.key
```

```
chmod 600 root/root.key

openssl req -new -x509 -key root/root.key -sha256 -days 3650 \
  -subj '/C=BY/O=Lab PKI/CN=Lab Root CA pathlen0' \
  -addext 'basicConstraints=critical,CA:TRUE,pathlen:0' \
  -addext 'keyUsage=critical,keyCertSign,cRLSign' \
  -addext 'subjectKeyIdentifier=hash' \
  -out root/root.crt

cat > inter/inter.ext <<'EOF'
basicConstraints=critical,CA:TRUE,pathlen:0
keyUsage=critical,keyCertSign,cRLSign
subjectKeyIdentifier=hash
authorityKeyIdentifier=keyid:always
EOF

openssl genpkey -algorithm RSA -pkeyopt rsa_keygen_bits:3072 -out
inter/inter.key
chmod 600 inter/inter.key

openssl req -new -key inter/inter.key \
  -subj '/C=BY/O=Lab PKI/CN=Lab Intermediate CA pathlen-test' \
  -out inter/inter.csr

openssl x509 -req -in inter/inter.csr \
  -CA root/root.crt -CAkey root/root.key -CAcreateserial \
  -sha256 -days 1825 -extfile inter/inter.ext \
  -out inter/inter.crt

openssl x509 -req -in ../server/server.csr \
  -CA inter/inter.crt -CAkey inter/inter.key -CAcreateserial \
  -sha256 -days 365 -extfile ../server/server.ext \
  -out leaf.crt

openssl verify -CAfile root/root.crt -untrusted inter/inter.crt leaf.crt
```

```
.+......+.....+................+......+.........+...+..+......+..........
............+.........+......+..+.+...........+.............+.+.....+...
+++*.......+....+.....+.+...+...+........+.+.....+.++++++++++++++++++++++
.....++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
.....+........+.+..+....+......+..+.......+.....++++++++++++++++++++++++++
++++++++++++++++++++++++*.+.......+...........+.+......+...+...........
++++++++
Certificate request self-signature ok
subject=C = BY, O = Lab PKI, CN = Lab Intermediate CA pathlen-test
Certificate request self-signature ok
subject=C = BY, O = Lab PKI, CN = localhost
C = BY, O = Lab PKI, CN = Lab Root CA pathlen0
error 25 at 2 depth lookup: path length constraint exceeded
error leaf.crt: verification failed
```